

① Reaction Rate

1° Factors

1) Physical **state** of the reactants

→ collide 碰撞 丸 Homogeneous often faster

2) Reactant **concentrations**

接触多

3) Reaction **temperature** ↑而↑

4) Presence of a **catalyst** 催化剂

改变 pathway
机理
碰撞类型

- Since there are more molecules, more **collisions** occur.

- Kinetic energy of molecules is related to temperature.
- At higher temperatures, molecules move more quickly, increasing **numbers** of **collisions** and the **energy** the molecules possess during the collisions.

2° rate expression

(单位统一是 $M \cdot s^{-1}$)

$\frac{\Delta[\]}{\Delta t}$

k 单位可变

➤ **average** rate 平均速率

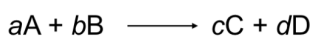
➤ **instantaneous** rate 瞬时速率

➤ **initial** rate 起始速率

stoichiometry

化学计量

- To generalize, then, for the reaction



方程的

$$\text{Rate} = -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t} = \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t}$$

3° rate law [order]

$$\text{Rate} = k[\][\]$$

rate constant

(temperature — dependent)

reaction 的 order 是每个反应物 order 的总和

• Heterogeneous reactions that involve solids are faster if the surface area is increased; i.e., a fine powder reacts faster than a pellet or tablet.

粉末

表面积

药片

NOTES

- Rate law are always determined experimentally.
- Reaction order is always defined in terms of reactant (not product) concentrations.
- The order of a reactant is **not related** to the stoichiometric coefficient of the reactant in the balanced chemical equation.

Chemical
Kinetics

② order Reaction (只有一个反应物)

1^o first order

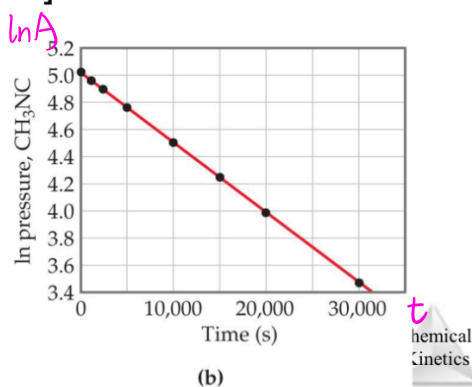
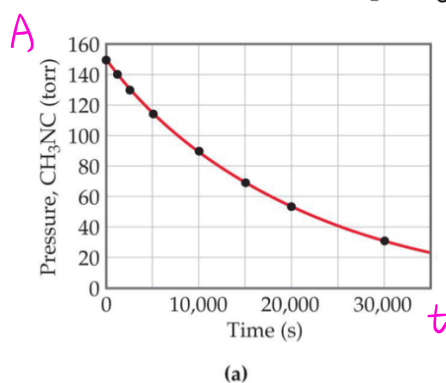
$$-\frac{\Delta[A]}{\Delta t} = k[A]$$

$$-\frac{\Delta[A]}{[A]} = k\Delta t$$

$$-\ln A = kt + C \quad C = -\ln A_0$$

$$\star -kt + \ln A_0 = \ln A$$

$$\star t_{1/2} = \frac{\ln 2}{k} \quad (\text{是一个常数})$$



2^o second order

$$-\frac{\Delta[A]}{\Delta t} = k[A]^2$$

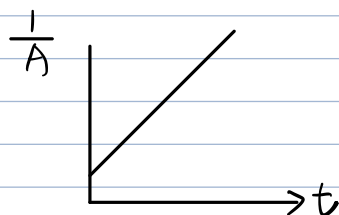
$$-\frac{\Delta[A]}{[A]^2} = k\Delta t$$

$$\frac{1}{[A]} = kt + C$$

$$\frac{1}{[A]} = kt + \frac{1}{[A_0]}$$

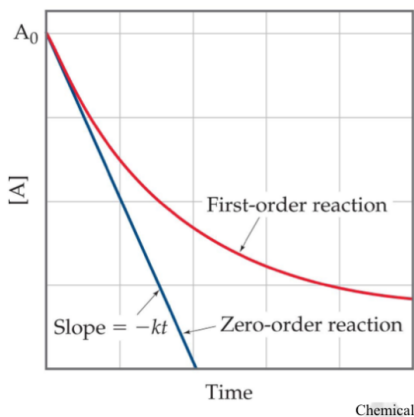
$$t_{1/2} = \frac{1}{k[A_0]}$$

相对于要求时间的初始浓度



3rd Zero order

$$\text{Rate} = k$$



③ Factors (更多从law的角度)

1. Temperature
2. Frequency of collisions
3. Orientation of molecules 方向

★ Energy needed for the reaction to take place (activation energy)
活化能

- The **rate constant** is **temperature-dependent**: it increases as temperature increases.

3×10^{-3}

Frequency of Collisions

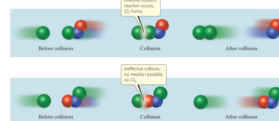
- The **collision model** is based on the **kinetic molecular theory** (分子动理论).
- Molecules must collide to react.
- If there are more collisions, more reactions can occur.
- So, if there are more molecules, the reaction is faster.
- Also, if the temperature is higher, molecules move faster, causing more collisions and a higher rate of reaction.

Chemical Kinetics

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Orientation of Molecules

- Molecules can often collide without forming products.
- Aligning molecules properly can lead to chemical reactions.
- Bonds must be broken and made and atoms need to be in proper positions.



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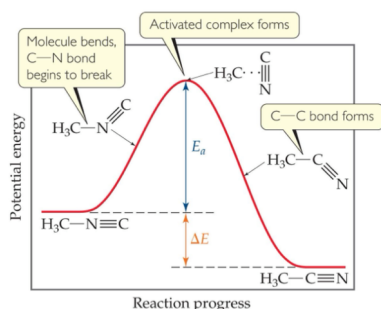
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energy barrier 能垒

Transition State (Activated Complex) 过渡态, (unstable)

Reaction Progress

- Plots are made to show the energy possessed by the particles as the reaction proceeds.
- At the highest energy state, the transition state is formed.
- Reactions can be **endothermic or exothermic** after this.

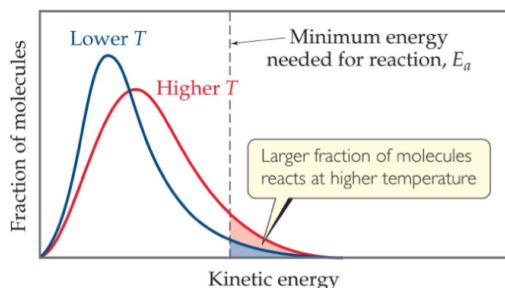


Chemical

无关吸放热

Distribution of the Energy of Molecules

- Gases have an average temperature, but each individual molecule has its own energy.
- At higher temperature, more molecules possess the energy needed for the reaction to occur.

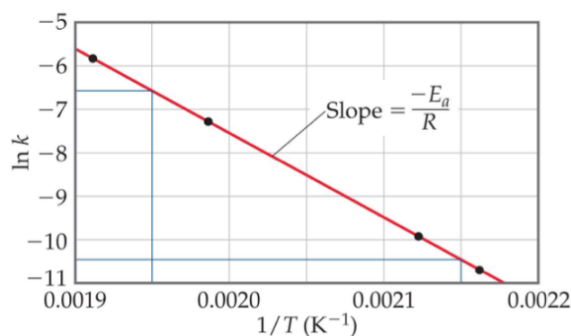


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The Relationship Between Activation Energy & Temperature

- Arrhenius** noted relationship between activation energy and temperature: $k = Ae^{-E_a/RT}$
- Activation energy can be determined graphically by reorganizing the equation: $\ln k = -E_a/RT + \ln A$
- A** is the frequency factor (a constant)



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$$\ln \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Chemical Kinetics

④ Elementary reaction 基元反应

每一个基元反应可以直接其 rate = k[]^{系数→计量数}

有三类

Molecularity

Table 14.3 Elementary Reactions and Their Rate Laws

Molecularity	Elementary Reaction	Rate Law
Unimolecular	$A \longrightarrow \text{products}$	$\text{Rate} = k[A]$
Bimolecular	$A + A \longrightarrow \text{products}$	$\text{Rate} = k[A]^2$
Bimolecular	$A + B \longrightarrow \text{products}$	$\text{Rate} = k[A][B]$
Termolecular	$A + A + A \longrightarrow \text{products}$	$\text{Rate} = k[A]^3$
Termolecular	$A + A + B \longrightarrow \text{products}$	$\text{Rate} = k[A]^2[B]$
Termolecular	$A + B + C \longrightarrow \text{products}$	$\text{Rate} = k[A][B][C]$

→ 只针对基元反应

The **molecularity** of an elementary reaction tells how many molecules are involved in that step of the mechanism.

Chemical

Termolecular 反应得最慢且 rare

rate-determining / limiting step 决速步

整体总反应 rate = 决速步 (基元反应) rate

What is Required of a Plausible Mechanism?

- The *rate law* must be able to be derived from the rate-determining step.
- The stoichiometry must be obtained when all steps are added up.
- Each step must balance, like any equation.
- All intermediates are made and used up.
- Any catalyst is used and regenerated.

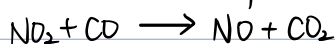
先生成再消耗 (其实相比过渡态是 stable)

中间体

先消耗再生成

Example

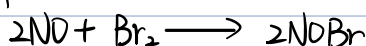
slow initial step



$$\text{Rate} = k[\text{NO}_2]^2$$

中间体 NO_3

fast initial step



$$\text{Rate} = k[\text{NO}]^2[\text{Br}_2]$$

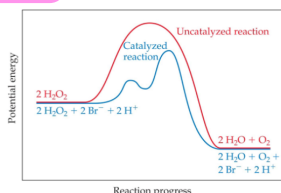
- Step 1: $\text{NO} + \text{Br}_2 \rightleftharpoons \text{NOBr}_2$
- Step 2: (rds) $\text{NOBr}_2 + \text{NO} \rightarrow 2\text{NOBr}$

$$k_1^+[\text{NO}][\text{Br}_2] = k_1^-[\text{NOBr}_2]$$

$$\text{rate} = k_2[\text{NOBr}_2][\text{NO}] = k_2 \frac{k_1^+}{k_1^-} [\text{NO}]^2[\text{Br}_2] = k[\text{NO}]^2[\text{Br}_2]$$

⑤ Catalysts

- Catalysts increase the rate of a reaction by decreasing the activation energy of the reaction.
- Catalysts change the mechanism by which the process occurs.



1) Homogeneous catalysts

2) Heterogeneous catalysts

3) Enzymes (catalysts in biological system)

酶

be xx order in [xxx]